P-192: labelled orbit extraction and the cost of caching

P-192: canonical cache ⟶ labelled collision equation
840 transitions;1512 point updates;10 planted extractions checked

Equal raw storage: five-point cache vs three ordinary states
Generic field-cost ratio >=3 before ideal sqrt(5) step reduction

Global order-five linear carrier requires four coordinates
Marked-subgroup-only actions and shared arithmetic remain outside this bound

2026-11-07, Round44. Constructed and independently checked: a multiwalk version of the saved order-five orbit cache, canonical scalar-label transport, and recovery from ten planted equal-orbit collisions. A separate cost derivation shows that equal raw state storage changes the evaluation: under the generic affine field-operation model, a five-point cache costs at least three times as much per walk transition as a negation-only baseline that uses that storage for three walks. Even granting the ideal sqrt(5) reduction in collision-search steps does not offset that field cost. This is a scoped cost result, not a universal lower bound on orbit algorithms or total runtime.

Target and prior state. Use standard P-192, p=2^192-2^64-1, r=0xFFFFFFFFFFFFFFFFFFFFFFFF99DEF836146BC9B1B4D22831, a=-3 and b=0x64210519E59C80E70FA7E9AB72243049FEB8DEECC146BC9B1. The scalar mu=407218087341981910240342834776973793515492198947424155465 4 has exact order5 modulo r. Round18 supplies the five-point representation and its total update. Round43 completed a batched isogeny point evaluator; this round checks the orbit-representative step rather than constructing another map. No isogeny endpoint or kernel is changed. The CM field remains K=Q(sqrt(D)), D=-241093790603361101225441612331139756649492725178968653595 15, with maximal endomorphism order.

Canonical representative and labels. Store T(R)=(R,mu*R,mu^2*R,mu^3*R,mu^4*R). For nonzero R all five x coordinates are distinct: equality would imply mu^(i-j)=+1 or-1 modulo r, impossible for distinct exponents of an odd-order subgroup. Choose the least x coordinate, rotate it first, and choose the sign giving its smaller y coordinate. This is a unique representative under the ten scalars +/-mu^j. If R=aP+bQ and the selected rotation/sign multiplier is z, its labels become (za,zb) modulo r. Infinity is handled separately.

Walk and extraction. A sixteen-entry table contains c_i*P+d_i*Q and its cached images. The table index is SHA256 of the canonical first point modulo16. Add the selected entry to every cached coordinate, then canonicalize and update labels. For equal canonical points with labels (a,b) and(c,d), Q=xP gives x=(a-c)/(d-b) modulo r whenever d-b is nonzero. Ten planted controls cover every rotation and sign and independently recover the fixture scalar20261107; each recovered value is checked by xP=Q. These are deliberately constructed collision controls, not discovery of a collision or an unknown-key search. Equal labels give a zero denominator and are rejected.

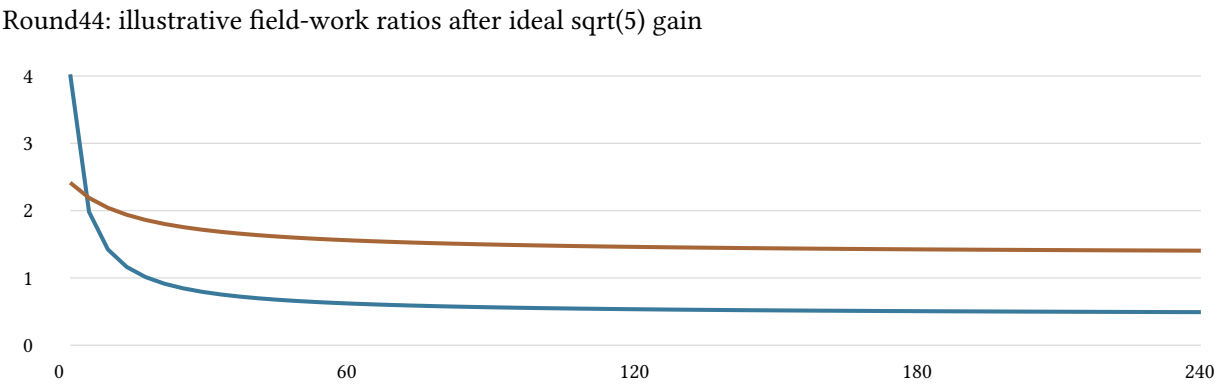
Finite verification. Frozen cached batch sizes were1,4,16, each run for8 transitions. Negation-only baselines used B and3B walkers with the same addition and inversion arithmetic. The independent checker replays840 labelled transitions and1512 individual point additions, checks canonicalization and labels, verifies all72 generic batch cost records, and checks the ten extraction equations. Identity, cancellation and doubling are separate branch controls. The short traces contain92 repeated-state records, all with identical labels and hence fruitless. Runs with overlapping initial fixtures are not independent samples; these counts are not evidence of a mixing or speed advantage. No restart or cycle record was discarded.

Generic arithmetic. A generic affine addition uses2M+S and one denominator inversion. Simultaneously inverting n denominators costs3(n-1)M+I, so a batch of n additions costs(5n-3)M+nS+I+6nA. A five-point cached batch of B transitions therefore uses(25B-3)M+5BS+I+30BA before canonicalization. The same-walker-count baseline uses(5B-3)M+BS+I+6BA. Canonicalization adds B+(m-1)*n_negative field negations/additions for an m-point cache, with m=1 for the baseline. Scalar-label operations, hashing, comparisons, setup and allocation are kept outside these field-operation counts. Initialization computes cached scalar images explicitly; its cost is not treated as a free new endomorphism.

Why equal walker counts can mislead. Write C=5*cM+cS and u=cI-3*cM. For an m-point cache and B walkers, per-transition MSI cost is m*C+u/B; for a one-point baseline at B walkers it is C+u/B. Under an ideal sqrt(m) quotient gain, caching wins this optimistic comparison exactly when cI>3*cM+sqrt(m)*B*C. A positive control with m=5,B=1,cM=cS=1,cI=80 gives107 versus83 per transition and passes that conditional test. Those weights are an arithmetic example, not measured P-192 timings.

Equal-storage theorem. Each affine coordinate and each scalar label occupies24 raw bytes on P-192. A five-point state with two labels uses288 bytes; an ordinary affine state with two labels uses96. Thus the same raw state storage holds3B baseline walkers. More generally, an odd m-point cache has2m+2 field-sized words versus4, giving kappa=(m+1)/2 baseline walkers per cached walker. Its per-transition costs satisfy C_cache-kappa*C_base=(m-kappa)*(5*cM+cS)>=0, where C_base=C+u/(kappa*B). This identity holds for all nonnegative MSI weights with positive baseline cost; it needs no inversion-dominance assumption. Since kappa>sqrt(m) for m>1, even the generous ideal collision gain leaves a field-work ratio at least(m+1)/(2*sqrt(m)). For m=5 the

bound is3/sqrt(5), approximately1.342. Field-addition accounting only strengthens this comparison for the recorded schedule. There are2480 exact rational-weight checks of the identity and inequality.



Blue: same walker count. Brown: equal raw storage. Ratio1 is parity.
x: inversion weight with M=S=1; B=1. Derived cost model, not timing or walk statistics.

Four-coordinate carrier limit. The standard companion carrier stores (R,mu*R,mu^2*R,mu^3*R), using mu^4=-1-mu-mu^2-mu^3. Its rotation matrix has superdiagonal ones and last row(-1,-1,-1,-1). Exact integer checks give C^5=I, 1+C+...+C^4=0, and the required marked eigenvector modulo r. Its raw state is240 bytes, enough for2.5 ordinary states. If its four coordinates are updated by generic additions, the same identity gives a per-transition factor at least2.5, even if obtaining the fifth point and canonicalizing are free. Dividing by ideal sqrt(5) leaves a field-work factor at least sqrt(5)/2, approximately1.118. This four-coordinate cost is derived, not an executed walk arm.

Dimension theorem and boundary. There is no nonidentity global order-five origin-fixing homomorphism on E^d for d<4. Indeed End^0(E^d)=M_d(K). The unique quadratic subfield of Q(zeta5) is real Q(sqrt5), so imaginary quadratic K has trivial intersection with it and Phi5 remains irreducible over K. A nonidentity A with A^5=I must therefore have Phi5 in its minimal polynomial, forcing d>=4. The displayed companion attains equality. This rules out a two-coordinate global finite-order linear carrier of that form. It does not exclude maps of order five only on the marked subgroup, nonlinear encodings, a different basis with special marked-coordinate behavior, or arithmetic shared between coordinates. The dimension lemma alone is not a bound on their costs.

Interpretation and next construction. The explicit cache and extraction equations work. Increasing the number of cached affine points, while using generic coordinatewise additions, does not supply the desired field-cost advantage at equal raw state storage. The next route must change that update model, use a genuinely cheap action on fewer stored coordinates, or extract scalar information without full orbit canonicalization. The three original field invariants do not themselves alter this cache accounting. This narrows the implementation route; it does not assert that all constructions based on those invariants are impossible.

Sources, limits and execution. Fruitless cycles and the need to account for their treatment are established concerns; see Bernstein, Lange and Schwabe, On the correct use of the negation map in the Pollard rho method, https://eprint.iacr.org/2011/003.pdf . Cyclotomic background: https://crypto.stanford.edu/pbc/notes/numberfield/cyclo.html . No novelty is claimed for Pollard equivalence classes or the companion representation. The sqrt(m) gain is a favorable collision-model assumption, not established for these short walks. Raw storage excludes compression, histories, tables and runtime objects. Common overheads and cycle-escape policies can change total runtime; no runtime bound is claimed. Scope preceded execution, local commands had120-second caps, no arithmetic checks failed and no cloud jobs or external state were changed. The report, two-dimensional cost identities, finite controls, scripts and inspected PDF are retained with hashes.